Schur construction panel: launch and custody

Running experiment snapshot · 2026-11-01T20:42:23.042429+00:00

The frozen Schur panel is executing on its original producer. This observation contains 17 complete context rows: 15 checked Schur constructions and 2 certified leading-block obstructions. These are prefix observations. No complete-panel or export result exists at this snapshot. The postprocessor follows this same invocation.

Complete prefix at the recorded observation: 17 of 642 contexts

Blue: 15 Schur identities; amber: 2 leading-block obstructions

Both are checked branches. This is an incomplete running panel.

Raw source: runs/cases.jsonl, first 245249835 bytes. The append-only prefix hash and complete row IDs are in snapshot.json. This graph counts completed branches, not statistical success estimates or a full recovery result.

Full scope. Each arm retains all 81 source contexts, using their balanced leading block, and adds 240 synthetic contexts. Synthetic cases cross primes 7, 13, 19, 43; dimensions 2, 3, 4, 7, 12; cuts 1, floor(N/2), N-1 with duplicates removed; and regular, zero-Schur, singular-Schur, invertible-full with blocked-leading, and zero-matrix types. Two arms give 642 rows. All leading obstructions remain in scope, including 96 synthetic rows whose full matrix is invertible despite a singular leading block.

Checks. Exact reference elimination supplies a two-sided leading inverse or a nonzero leading nullvector. With an inverse, check the full Schur displacement, width s+2, all nested source and oracle counts, and both product directions through the Schur wrapper and generated companion operator. With a blocked inverse, generation and application must raise the specific obstruction and retain partial work. That obstruction does not establish full-matrix singularity. Equal-wrap, output, count and generator controls are fixed before data.

Custody. Predata commit f5cac20. Protocol hash: 7403509a477e478f03427db64fd4c864e8e62ea99dba3f5d84f8bcd249204480. All 642 execution-row seeds matched before data. Eleven source files and one public input archive are locked. One fresh timeout 3600 invocation; no restart or extension. Producer PID 2104676, session 68027; native timeout PID 2104675. Postprocessor PID 2106367, session 68331. launch.json records start ticks and a deadline estimate; the existing native timeout controls the actual deadline.

Costs. Generator construction schedules 2s+6 full-source applications and s+2 leading solves per direction. The reference inverse oracles record dense dot-product terms and output reductions separately from source and native work. Those calls are not free or evidence of fast inversion. No clock, CPU or RSS comparison is claimed. All failures and partial rows are retained.

Next action. Follow the existing processes. After producer exit 0, the postprocessor runs the frozen streaming analysis and full-file export check once. A missing owner without an exit receipt is an interruption, not a reason to rerun. After terminal checks, render and inspect a fresh final report and seal all raw records.

Derived next adapters: RHS and primal/dual lifting

Written and syntax-checked only. No adapter numerical trial ran in this round. Source: companion-schur-lift/lift.py; proof: companion-schur-lift/DERIVATION.md. These files do not alter the running Schur instrument.

For $A=[[E,B],[C,D]]$ with E invertible, $S=D-CE^{-1}B$, and $b=[b1;b2]$, set $h=b2-CE^{-1}b1$. Given any trailing vector z, define $x=[E^{-1}(b1-Bz);z]$. Then the exact residual is $Ax-b=[0;Sz-h]$. Thus a checked Schur solution lifts to a full-system solution.

For any trailing v, define $w=[-E^{-t}C^{t}v;v]$. Then $w^{t}A=[0,v^{t}S]$ and $w^{t}b=v^{t}h$. In particular, $v^{t}S=0$ and $v^{t}h=1$ produce a normalized full-system exclusion witness. The existing separate feature annihilation check remains necessary; a Gram-null identity alone does not establish that feature-level condition.

Each adapter uses one full-source application and one leading-block solve. Additional scalar work is m subtractions for RHS reduction and n for either lift, where $n=\dim(E),m=\dim(S)$. Dual lifting uses the transpose direction. The helper records those calls through the Schur object's counters and lets LeadingBlockObstruction propagate unchanged.

The next adapter gate must check both residual identities, normalized dual witnesses, changed RHS/sign controls and complete nested costs, after the current Schur construction panel is reconciled. Then a recursive backend must replace the explicit leading-block reference oracles. No recursive solver or new full ECDLP recovery is claimed by these unexecuted adapters.